

Reviewer-Auditable Reproducibility Without Transcript Release: An Aggregate-Only Framework for Sensitive Speech AI Evaluation

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Abstract

Reproducible evaluation is difficult when AI systems are tested on sensitive speech evidence. Raw audio, transcripts, row identifiers, reviewer notes, response sheets, model hypotheses, and runtime logs may contain private operational content, yet reviewers still need to inspect whether reported claims are supported by traceable evidence. This paper proposes aggregate-only reproducibility as a responsible-AI framework for sensitive speech evaluation. The framework separates public aggregate artifacts from local-only transcript-bearing materials and links each paper-facing claim to evidence layers, artifact manifests, validation gates, operation records, privacy classes, and explicit release boundaries.

We develop the framework through a CDS-ASR case study in high-stakes speech-to-decision evaluation. The case contains four distinct evidence layers: a 258-row ASR test split for model-comparison context, selected-300 provenance outputs for enriched audit-surface construction, 30 human-reviewed audit rows yielding 90 clustered model assessments for decision-change evaluation, and aggregate policy replay for recovery analysis. Instead of releasing raw speech or transcript-bearing row content, the case provides reviewer-visible auditability through aggregate tables, generated figures, claim registries, artifact manifests, row-clustered uncertainty summaries, leave-one-row-out sensitivity summaries, and consistency checks.

The contribution is claim-evidence stewardship. Aggregate-only reproducibility does not provide full row-level external reproduction and does not claim legal compliance, privacy completion, or deployment readiness. It offers a bounded audit mode for sensitive speech AI studies whose evidence can be made reviewer-auditable without public transcript release.

Keywords: trustworthy AI; reproducible evaluation; sensitive speech data; artifact manifest; claim-evidence alignment; responsible AI audit.

1 Introduction

Sensitive speech AI research often cannot release raw audio, raw transcripts, row identifiers, reviewer notes, response sheets, model hypotheses, or transcript-bearing runtime logs. These materials can contain private operational speech content and contextual details that remain subject to institutional, contractual, ethical, or safety constraints. At the same time, reviewers need to evaluate whether empirical claims are traceable, scoped, and supported.

This paper asks how reviewers can audit empirical claims when row-level transcript release is not appropriate. The answer proposed here is aggregate-only reproducibility: a bounded reproducibility mode that exposes aggregate evidence, artifact manifests, validation gates, operation records, claim registries, and release-boundary decisions while keeping transcript-bearing evidence local-only.

The paper uses the CDS-ASR project as a case study. The technical details of CDS-ASR and CEIS belong to the companion speech-evaluation manuscript. Here, CDS-ASR provides an example

of a sensitive speech AI evaluation where reviewer-visible auditability must be achieved without raw transcript release.

2 Background

Reproducibility in AI evaluation. AI evaluation commonly relies on public datasets, executable code, fixed metrics, and artifact checklists. Sensitive speech settings complicate this model because the evaluation evidence may include personally revealing acoustic or transcript-bearing content.

Sensitive speech data. Speech data can contain identity cues, operational details, medical or financial information, emotional content, and contextual signals that remain sensitive even after transcription. Releasing row-level examples can therefore create privacy and governance risks.

Claim-evidence alignment. For sensitive AI evaluation, the paper claim should specify its evidence layer, statistic, scope, limitation, privacy class, and validation status. This turns limitations into a claim-evidence alignment discipline rather than a defensive caveat. The framework is compatible with documentation-oriented responsible-AI practice such as dataset documentation [1] and risk-management framing [2], while making no legal-compliance claim.

Audit trails and operation records. Operation records preserve process-level reproducibility: what generator or validator produced an artifact, from which source inputs, under which version and environment, with which checksum. They do not expose local-only content, but they make the aggregate evidence trail inspectable.

3 Aggregate-Only Reproducibility Framework

We propose a five-layer framework for sensitive speech AI evaluation.

3.1 Layer 1: Evidence Boundary

The evidence boundary defines which materials are public aggregate artifacts and which remain local-only sensitive evidence. Figure 2 maps the boundary.

3.2 Layer 2: Aggregate Artifact Layer

The aggregate artifact layer releases transcript-free summary tables, metric outputs, validation summaries, manifest rows, and figure inputs. It does not release raw transcripts, row identifiers, reviewer notes, response sheets, model hypotheses, or transcript-bearing logs.

3.3 Layer 3: Claim Registry

The claim registry maps every paper-facing claim to artifact path, statistic, scope, limitation, privacy class, and validation status. Figure 3 shows the claim-to-evidence flow.

Five-layer aggregate-only reproducibility framework

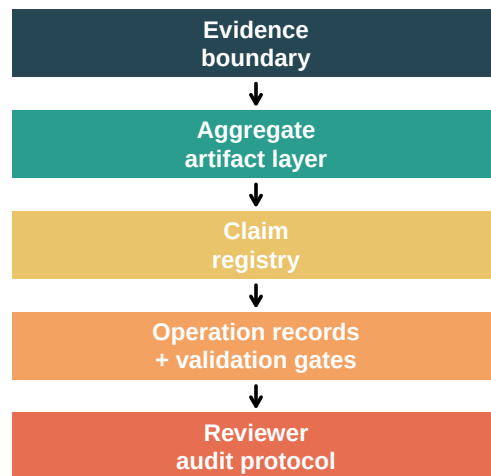


Figure 1: Five-layer aggregate-only reproducibility framework. The figure is generated by `paper_4b_aggregate_reproducibility/r/build_fig1_framework_layers.R` as a conceptual framework derived from the manuscript protocol.

Evidence boundary map



Reviewer-visible auditability is aggregate-only; row-level transcript evidence remains governed locally.

Figure 2: Evidence boundary map separating public aggregate artifacts from local-only sensitive evidence. Generated by `build_fig2_evidence_boundary.R`.

Claim registry flow



Every claim is routed through aggregate evidence and an explicit boundary.

Figure 3: Claim registry flow from paper claim to artifact, statistic, scope, limitation, privacy class, and validation status. Generated by `build_fig3_claim_registry_flow.R`.

3.4 Layer 4: Operation Records And Validation Gates

Operation records and validation gates provide process-level audit trails without exposing sensitive content. They include artifact manifests, evidence-chain readiness checks, consistency checks, publishable-evidence summaries, and leak checks.

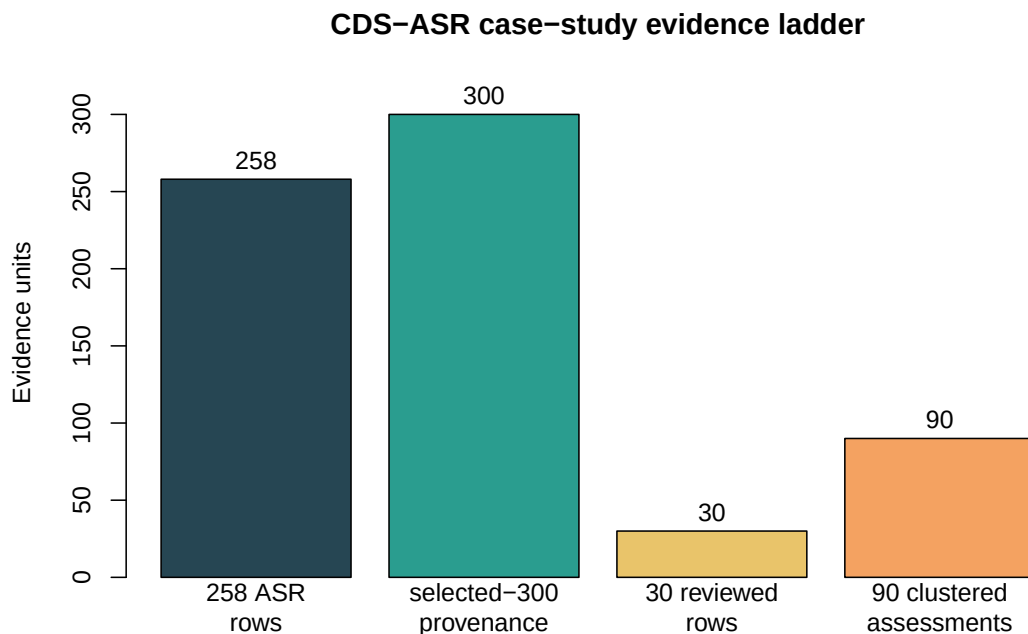
3.5 Layer 5: Reviewer Audit Protocol

The reviewer audit protocol specifies what reviewers can inspect: artifact completeness, claim-evidence alignment, consistency checks, metric definitions, scope boundaries, and R-based figure regeneration. It also specifies what reviewers cannot inspect in a public package: raw transcript content, row-level qualitative interpretation, and hidden local-only sensitive evidence.

4 Case Study: CDS-ASR

CDS-ASR is used here only as a case study for aggregate-only reproducibility. The companion Paper 4-a carries the speech-evaluation method and CEIS technical claims.

The case has four evidence layers: a 258-row ASR split, selected-300 enriched high-stakes provenance outputs, 30 human-reviewed audit rows, and 90 model-row assessments clustered within those 30 rows. Figure 4 shows the case-study evidence ladder.



Case evidence is aggregate-only; selected-300 is enriched and 90 assessments are clustered.

Figure 4: CDS-ASR case-study evidence ladder. Generated by `paper_4b_aggregate_reproducibility/r/build_fig5_case_study_evidence_ladder.R`.

Predictor comparison and policy replay appear as aggregate outputs. Raw audio, raw transcripts, row identifiers, reviewer notes, response sheets, model hypotheses, and transcript-bearing logs

remain local-only. This makes the case appropriate for demonstrating reviewer-auditable aggregate evidence while preserving sensitive material.

5 Artifact And Claim-Evidence Protocol

The protocol combines artifact manifests, claim registries, evidence-chain consistency summaries, publishable-evidence completion summaries, roadmap completion summaries, operation records, validation gates, and privacy classes.

Table 1: Public, aggregate-only, and local-only evidence classes.

Class	Examples	Reviewer use
Public aggregate	Summary tables, generated figures, validation summaries	Check reported aggregate claims
Public source/config	Figure scripts, manifests, metric definitions	Regenerate public figures and inspect definitions
Controlled aggregate supplement	Extended aggregate tables and sensitivity summaries	Inspect scoped uncertainty and consistency
Local-only sensitive	Raw audio, transcripts, row IDs, reviewer notes, response sheets, hypotheses	Not public; governed locally or institutionally

Table 2: Claim-evidence alignment schema.

Field	Purpose
Claim	Paper-facing statement being supported
Artifact path	Aggregate artifact used as evidence
Statistic	Reported value or validation status
Scope	Evidence layer and population boundary
Limitation	Claim-evidence alignment boundary
Privacy class	Public aggregate, controlled aggregate, or local-only
Validation status	Gate, check, or operation record supporting use

The core case-study artifacts include `70_experiments/runs/postdoc_evidence_chain_2026_05_25/artifact_manifest.tsv`, `70_experiments/runs/postdoc_evidence_chain_2026_05_25/claim_registry.tsv`, `70_experiments/runs/postdoc_evidence_chain_2026_05_25/evidence_chain_consistency_summary.json`, `70_experiments/runs/postdoc_evidence_chain_2026_05_25/publishable_evidence_completion_summary.json`, and `70_experiments/runs/postdoc_evidence_chain_2026_05_25/postdoc_roadmap_completion_summary.json`.

6 Reviewer-Auditable Boundary

Reviewer-visible auditability is not the same as full row-level reproducibility. Reviewers can inspect aggregate artifacts, figures, tables, manifests, validation gates, claim registries, and release-boundary statements. They cannot independently inspect raw transcript cases in the public package.

Table 3: Artifact manifest fields and purpose.

Field	Purpose
Path	Locates the aggregate artifact
Role	Explains paper-facing function
Generator	Identifies the script or process that produced it
Source inputs	Defines upstream aggregate evidence
Checksum	Supports integrity checks
Source commit	Links artifact to repository state
Environment	Records runtime and package context
Privacy class	Confirms release boundary

Reviewer auditability map

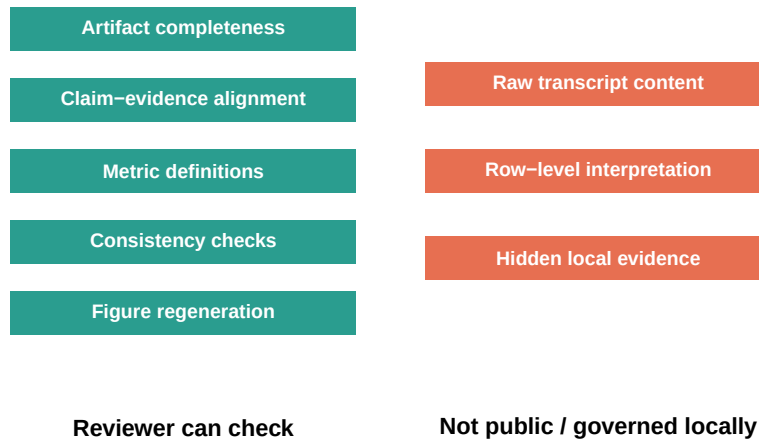


Figure 5: Reviewer auditability map showing what can be checked and what remains local-only or institutionally governed. Generated by `build_fig4_artifact_audit_map.R`.

Table 4: Reviewer audit checklist.

Reviewer check	Public evidence	Boundary
Artifact completeness	Manifests and aggregate package list	Does not expose raw rows
Claim-evidence alignment	Claim registry	Claims remain scoped
Metric definitions	Method summary and metric audit	Raw transcripts not released
Consistency checks	Evidence-chain summaries	Aggregate validation only
Figure regeneration	R scripts and aggregate inputs	R scripts cannot access sensitive inputs

Table 5: CDS-ASR case-study claim boundary table.

Case evidence	What it supports	Boundary
258 ASR rows	Model-comparison context	Not deployment safety evidence
selected-300 provenance	Enriched audit surface	Not prevalence-preserving
30 reviewed rows	Human-reviewed audit layer	Single-expert audit
90 model assessments	Clustered metric evidence	Not independent calls
Aggregate replay	Policy-replay auditability	Retrospective, not live deployment

7 Discussion

Aggregate-only reproducibility is useful for sensitive speech AI domains where open row-level release is inappropriate: medical speech, fraud calls, psychotherapy speech, law-enforcement-adjacent speech data, and high-stakes conversational AI evaluation. The framework gives reviewers a structured way to inspect the public evidence layer without requiring public release of transcript-bearing materials.

The framework also clarifies what aggregate auditability can and cannot support. It can support claim-evidence alignment, artifact completeness checks, metric-definition checks, consistency checks, and figure regeneration from aggregate inputs. It cannot support independent qualitative review of hidden local-only transcript rows.

8 Limitations And Ethics

Aggregate-only reproducibility is weaker than row-level reproducibility. Reviewers cannot independently inspect raw transcript cases. The framework depends on honest local execution, institutional controls, and accurate manifests. The CDS-ASR case study is one project, not universal validation of the framework.

This paper does not claim legal compliance and does not replace IRB review, data-use agreements, retention policies, encryption, or access control. It provides a claim-evidence governance framework for reviewer-auditable aggregate evidence under sensitive-data constraints.

A Appendix: Aggregate Artifacts

The public package should include aggregate artifacts only. Transcript-bearing materials remain local-only.

- Artifact manifest: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/artifact_manifest.tsv`
- Claim registry: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/claim_registry.tsv`
- Evidence-chain consistency: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/evidence_chain_consistency_summary.json`
- Publishable evidence completion: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/publishable_evidence_completion_summary.json`

- Roadmap completion: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/postdoc_roadmap_completion_summary.json`
- Paper artifact manifest: `80_semantic_risk_asr/paper/artifact_manifest.tsv`

References

- [1] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daume III, and Kate Crawford. Datasheets for datasets, 2018. Documentation background.
- [2] National Institute of Standards and Technology. Artificial intelligence risk management framework, 2023. Governance context; not cited as legal compliance evidence.